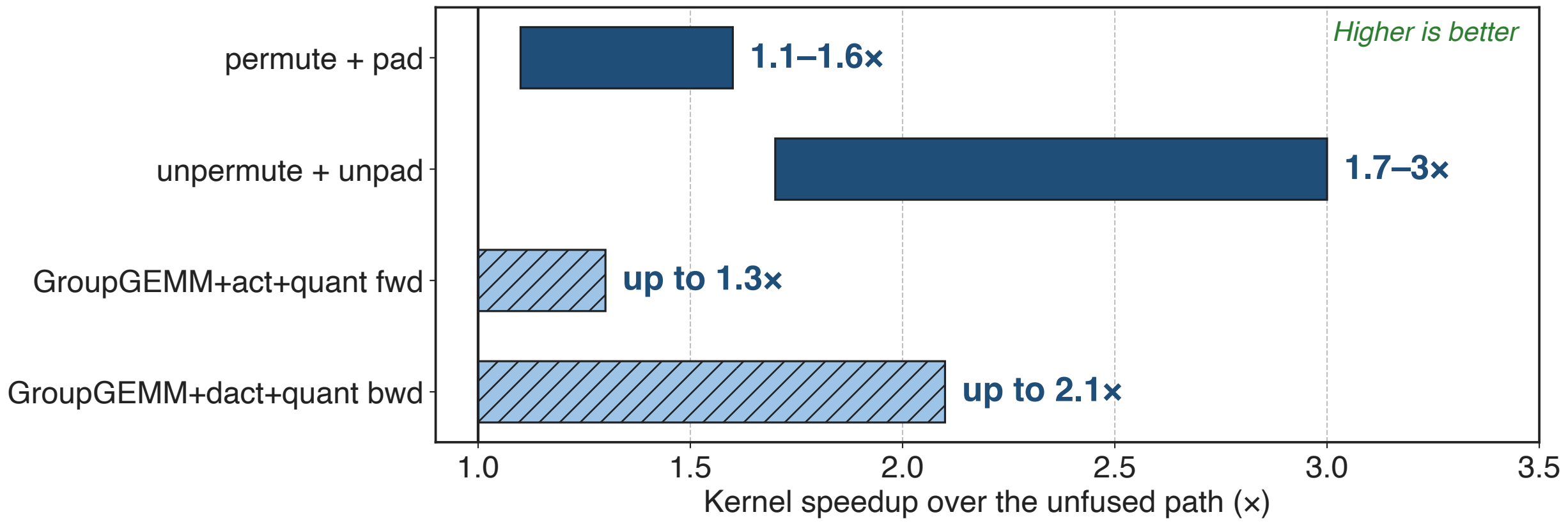
Higher is better
permute + pad
1.1–1.6×
unpermute + unpad
1.7–3×
GroupGEMM+act+quant fwd
up to 1.3×
GroupGEMM+dact+quant bwd
up to 2.1×
1.0
1.5
2.0
2.5
3.0
3.5
Kernel speedup over the unfused path (×)